

# A Computational Framework for Emergent Multi-Sector Physical Structures from Discrete Energy Fields

Toward an Oscillatory Substrate Interpretation

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## Abstract

We present a discrete computational framework in which multiple physical behaviours emerge from a common underlying structure based on graph dynamics.

Starting from filamentary configurations extracted from scalar field variations, we construct graph-based representations that support distinct effective regimes. These include wave-like propagation analogous to Schrödinger dynamics, fermionic behaviour governed by a Dirac-like operator, gauge-like transverse modes arising from edge-based Laplacians, and geometry–source coupling induced by collective density fields.

Each sector is analysed numerically across multiple parameter regimes and independent realizations. The results show consistent and robust patterns, including linear dispersion relations, spectral pairing structures, transverse mode dominance, and stable correlations between effective geometric quantities and source-like fields.

Importantly, these behaviours are not imposed through continuum equations but arise from the intrinsic properties of the discrete system. This suggests that well-established physical equations may be interpreted as effective descriptions emerging from a common substrate.

While no direct equivalence with known physical theories is claimed, the framework provides a unified computational setting in which multiple physical regimes coexist and interact, offering a potential avenue toward an emergent understanding of fundamental dynamics.

## 1 Introduction

Understanding whether the diverse laws of physics can arise from a common underlying structure remains one of the central challenges of theoretical physics. While quantum mechanics, gauge theories, and general relativity each provide highly successful descriptions within their respective domains, their unification at a fundamental level remains an open problem.

The present work does not aim to replace or reinterpret these established theories. Instead, it proposes a computational framework in which several of their characteristic behaviours emerge from a shared discrete substrate. In this sense, the objective is not to redefine known physics, but to explore whether its structures can be recovered as effective regimes of a more elementary organization.

More precisely, we construct a graph-based framework derived from filamentary structures extracted from scalar field variations. On this discrete support, we define operators that give rise to distinct dynamical sectors, including:

- wave-like propagation analogous to Schrödinger dynamics,
- spectral structures associated with Dirac-like operators,

- transverse modes arising from edge-based Laplacians reminiscent of gauge fields,
- and geometry–source couplings emerging from collective density fields.

These behaviours are not imposed through continuum equations. Rather, they arise from the intrinsic properties of the discrete structures obtained from the initial data. This perspective suggests that well-established physical laws may be interpreted as effective descriptions of a common underlying organization, instead of independent fundamental ingredients.

It is important to emphasize that this work does not claim equivalence with existing physical theories. The framework remains dimensionless and computational, and no direct identification with experimentally measured quantities is established at this stage. The goal is instead to demonstrate the coexistence of multiple physically meaningful regimes within a single, unified construction.

Finally, this work is presented with a deliberate emphasis on clarity and reproducibility. The associated codebase provides a fully deterministic implementation of the pipeline, allowing all reported results to be independently verified.

This work is not intended to replace or compete with established physical theories, but rather to propose a unifying perspective in which their respective successes can be understood as emerging from a common underlying structure.

The remainder of this paper is organized as follows. Section 2 introduces the discrete scalar fields and their role as input data. Section 3 presents filament extraction and graph construction. Section 4 defines the discrete operators on the filament graph. Sections 5–8 analyse the emergent Schrödinger, Dirac, gauge, and geometry–source sectors. Section 9 introduces the minimal dynamical structure associated with spectral pair oscillations. Section 10 presents the unified structural formulation of the framework. Section 11 addresses code availability and reproducibility. Sections 12 and 13 provide discussion and conclusion, respectively. Supporting material is collected in the appendices.

## 2 Raw Data and Discrete Fields

The framework operates on discrete scalar fields defined on a four-dimensional lattice. Each configuration is represented as a NumPy array of the form:

$$E(x, y, z, t),$$

where  $(x, y, z, t)$  denote discrete spatial and temporal indices on a regular grid.

In the present implementation, the lattice size is fixed to  $12 \times 12 \times 12 \times 12$ , as reflected in the dataset filenames. These scalar fields represent local energy-like quantities obtained from prior numerical simulations.

We denote by  $\nabla E$  the discrete gradient of the field, defined using periodic finite differences along each lattice direction. This quantity will play a central role in the identification of structurally relevant regions in the next section.

### 2.1 Origin of the Data

The input configurations are generated from Monte Carlo simulations based on an  $SU(3)$  lattice model. The precise implementation used in this work corresponds to a modified version of a standard lattice Monte Carlo evolution, producing local scalar observables extracted from the underlying gauge configurations.

The scalar field  $E$  should therefore be understood as a derived quantity capturing local structural variations of the system, rather than a fundamental physical field.

No specific physical interpretation is imposed at this stage. The framework only assumes that  $E$  is a scalar field defined on a discrete four-dimensional domain.

## 2.2 File Structure and Naming Convention

Each configuration is stored as a NumPy file (`.numpy`) following a consistent naming convention:

`local_energy_betaXX.XX_seedY_hot_fx0_fz0_V12x12x12x12... .numpy`

The main parameters encoded in the filenames are:

- $\beta$ : coupling parameter of the underlying simulation,
- `seed`: index of the random initialization,
- lattice size: here fixed to  $12^4$ ,
- additional tags indicating preprocessing variants (e.g. `fill`).

These files are provided in the repository under:

`data/raw_maps/`

and constitute the only external input required to reproduce all results presented in this work.

## 2.3 Role in the Framework

All subsequent constructions — including filament extraction, graph building, and the definition of effective operators — are derived deterministically from these scalar fields.

Importantly, the framework does not introduce any additional dynamical assumptions beyond the information already encoded in  $E$ . Instead, it investigates how structural properties of the field, such as local variations and connectivity patterns, give rise to emergent behaviours at larger scales.

In this sense, the scalar field serves as a minimal substrate from which multiple effective regimes are constructed.

## 2.4 Reproducibility

The entire pipeline is deterministic once the input fields are fixed. Given the provided dataset, all results reported in this paper can be reproduced by executing the public codebase without any manual intervention.

This ensures full transparency and allows independent verification of all numerical observations.

# 3 Filament Extraction and Graph Construction

## 3.1 Gradient-Based Filament Identification

The first structural layer of the framework consists in identifying regions of significant local variation within the scalar field  $E(x, y, z, t)$ .

We define the discrete gradient norm as:

$$G(x) = \|\nabla E(x)\|, \tag{1}$$

where the gradient is computed using periodic finite differences along each lattice direction.

To extract meaningful structures, we normalize  $G$  using a z-score transformation:

$$z_G(x) = \frac{G(x) - \mu_G}{\sigma_G}, \quad (2)$$

where  $\mu_G$  and  $\sigma_G$  denote the mean and standard deviation of  $G$  over the lattice.

We then define a critical field:

$$C(x) = \max(z_G(x), 0)^2, \quad (3)$$

which emphasizes high-gradient regions while suppressing background fluctuations.

A Gaussian smoothing is applied:

$$C_\sigma = \mathcal{G}_\sigma * C, \quad (4)$$

where  $\mathcal{G}_\sigma$  denotes a Gaussian kernel with width  $\sigma$ .

Finally, filamentary regions are defined by a high-percentile threshold:

$$\Omega = \{x \mid C_\sigma(x) \geq \text{Percentile}_p(C_\sigma)\}. \quad (5)$$

In practice,  $p = 99\%$  is used, isolating the most significant structural variations.

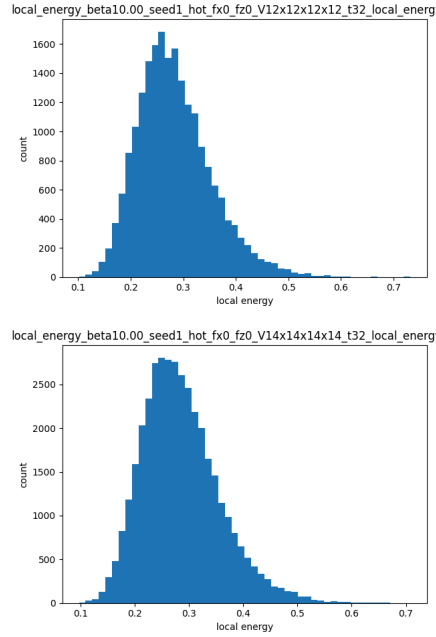


Figure 1: Distribution of gradient norms  $\|\nabla E\|$  used to define the filament extraction threshold.

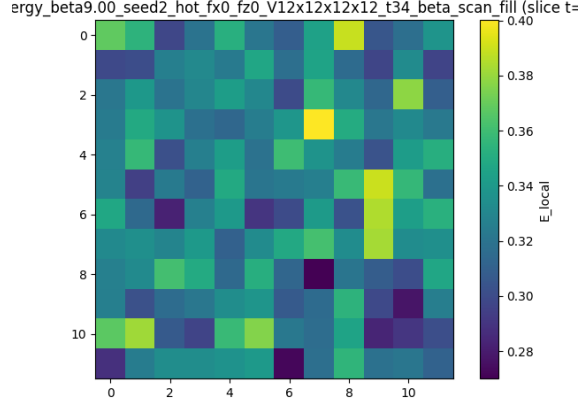


Figure 2: Representative slice of the scalar field highlighting localized variations used in the filament extraction stage.

### 3.2 Graph Construction from Filament Structures

The set  $\Omega$  defines a subset of lattice sites interpreted as a discrete filamentary structure.

We construct an undirected graph  $G = (V, E)$  as follows:

- Each point  $x \in \Omega$  defines a node  $v \in V$ ,
- Edges are defined between nearest neighbours on the lattice,
- Periodic boundary conditions are applied in all directions.

Formally, for two nodes  $u, v \in V$ , an edge  $(u, v) \in E$  exists if:

$$v = u + \hat{e}_\alpha \quad \text{for some lattice direction } \alpha, \quad (6)$$

with periodic wrapping.

To avoid double-counting, only forward neighbour directions are considered during construction.

Each node carries the following attributes:

- spatial coordinates  $(x, y, z, t)$ ,
- critical field value  $C_\sigma(x)$ ,
- degree (number of connected edges).

This construction yields a sparse graph encoding the backbone of high-gradient structures within the scalar field.

### 3.3 Structural Interpretation

The resulting graph represents a discrete skeleton of the underlying field.

Importantly:

- no additional dynamics are introduced at this stage,
- the graph is entirely determined by local variations of  $E$ ,

- all subsequent physical behaviours emerge from this structure.

This step therefore defines the fundamental substrate of the framework, on which all operators and emergent dynamics are constructed.

The graph constructed in this way provides the discrete substrate on which all operators and emergent dynamics are defined. We now introduce the fundamental operators acting on this structure.

## 4 Discrete Operators on the Filament Graph

### 4.1 Graph Representation

From the filament extraction step, we obtain an undirected graph  $G = (V, E)$ , where:

- $V$  is the set of nodes corresponding to selected lattice sites,
- $E$  is the set of edges defined by nearest-neighbour connectivity.

This graph serves as the fundamental discrete structure on which all subsequent operators are defined.

### 4.2 Adjacency and Degree

We define the adjacency matrix  $A \in \mathbb{R}^{N \times N}$  as:

$$A_{ij} = \begin{cases} 1 & \text{if nodes } i \text{ and } j \text{ are connected,} \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

The degree of a node  $i$  is given by:

$$d_i = \sum_j A_{ij}. \quad (8)$$

The diagonal degree matrix is:

$$D = \text{diag}(d_1, \dots, d_N). \quad (9)$$

### 4.3 Graph Laplacian

The combinatorial graph Laplacian is defined as:

$$L = D - A. \quad (10)$$

We also consider the normalized Laplacian:

$$L_{\text{norm}} = I - D^{-1/2} A D^{-1/2}, \quad (11)$$

where  $I$  is the identity matrix.

The normalized Laplacian plays a central role in defining diffusion-like and wave-like processes on the graph.

#### 4.4 Incidence Matrix and Edge Space

To describe edge-based dynamics, we introduce the incidence matrix  $B \in \mathbb{R}^{N \times M}$ , where  $M = |E|$ . Each column of  $B$  corresponds to an oriented edge  $(u, v)$  and is defined as:

$$B_{ie} = \begin{cases} -1 & \text{if node } i = u, \\ +1 & \text{if node } i = v, \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

The choice of orientation is arbitrary but fixed.

The incidence matrix allows us to define discrete differential operators on the graph:

- $B^T$  acts as a discrete gradient,
- $B$  acts as a discrete divergence.

#### 4.5 Edge Laplacian

The edge Laplacian is defined as:

$$L_e = B^T B. \quad (13)$$

This operator acts on edge-based fields and encodes circulation and transverse structures. It will be central to the emergence of gauge-like dynamics.

#### 4.6 Dirac-like Operator

We construct a Dirac-like operator using the incidence matrix:

$$D = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}. \quad (14)$$

This operator acts on the combined space of nodes and edges.

Its key properties include:

- a symmetric spectrum around zero,
- pairing of eigenvalues  $(\pm E)$ ,
- a natural coupling between node and edge degrees of freedom.

This structure will be shown to generate fermion-like behaviour in the emergent dynamics.

#### 4.7 Effective Hamiltonian

To describe wave-like propagation on the graph, we define an effective Hamiltonian:

$$H = \alpha L_{\text{norm}} + \beta V, \quad (15)$$

where  $V$  is a diagonal potential derived from node degree:

$$V_{ii} = \frac{d_i}{\max_j d_j}. \quad (16)$$

This normalization ensures that the potential remains bounded between 0 and 1.

The parameters  $\alpha$  and  $\beta$  control the relative contribution of diffusion and potential effects.

This operator governs the Schrödinger-like dynamics explored in the next section.

## 4.8 Structural Role of Operators

All subsequent physical behaviours emerge from these discrete operators. Importantly:

- no continuum equations are assumed,
- all operators are constructed directly from graph structure,
- different physical regimes correspond to different operator choices.

This establishes a unified discrete foundation for the emergence of multiple physical behaviours.

## 5 Emergent Schrödinger Dynamics

### 5.1 Effective Hamiltonian on the Graph

We now consider wave-like dynamics on the filament graph using the effective Hamiltonian introduced in the previous section:

$$H = \alpha L_{\text{norm}} + \beta V, \quad (17)$$

where  $L_{\text{norm}}$  is the normalized graph Laplacian and  $V$  is a degree-based potential.

This operator defines the evolution of a complex-valued state  $\psi \in \mathbb{C}^N$  over the graph nodes.

### 5.2 Time Evolution

The dynamics are governed by a Schrödinger-type equation:

$$i \frac{d\psi}{dt} = H\psi. \quad (18)$$

Given the eigen-decomposition:

$$H = U \Lambda U^\dagger, \quad (19)$$

the solution can be written as:

$$\psi(t) = U e^{-i\Lambda t} U^\dagger \psi_0, \quad (20)$$

where  $\psi_0$  is the initial state.

### 5.3 Initial Condition

We consider an initial condition localized on a high-degree node:

$$\psi_0(i) = \begin{cases} 1 & \text{if } i = i_{\text{max}}, \\ 0 & \text{otherwise,} \end{cases} \quad (21)$$

where  $i_{\text{max}}$  is the node with maximum degree.

This choice probes the propagation of excitation from structurally central regions of the graph.



## 5.4 Observed Quantities

We analyse the evolution using the following observables:

- maximum probability:

$$\max_i |\psi_i(t)|^2, \quad (22)$$

- effective support:

$$S_{\text{eff}} = \frac{1}{\sum_i |\psi_i|^4}, \quad (23)$$

- Shannon entropy:

$$S = - \sum_i p_i \log p_i, \quad p_i = |\psi_i|^2, \quad (24)$$

- return probability:

$$|\langle \psi_0, \psi(t) \rangle|^2. \quad (25)$$

These quantities characterize localization, spreading, and coherence of the evolving state.

## 5.5 Spectral Structure and Dispersion

The eigenvalues of  $H$  define a spectrum that can be interpreted in terms of effective modes.

We define a mode index  $k$  and an effective wave number:

$$k_{\text{eff}} = \sqrt{k}. \quad (26)$$

Empirically, the eigenvalues exhibit a quadratic scaling:

$$E_k \sim k_{\text{eff}}^2, \quad (27)$$

consistent with a dispersion relation of the form:

$$E \sim k^2. \quad (28)$$

This behaviour is characteristic of diffusive or non-relativistic wave propagation.

dispersion\_summary

| beta        | alpha | beta_pot | dispersion_slope_a   | dispersion_intercept_E0 | dispersion_fit_r   | group_velocity_mode1 |
|-------------|-------|----------|----------------------|-------------------------|--------------------|----------------------|
| <b>9.0</b>  | 2.0   | 1.0      | 0.027296554130559942 | 0.09208481141608771     | 0.9589763101512024 | 0.054593108261119884 |
| <b>13.0</b> | 0.5   | 1.0      | 0.018033446221493013 | 0.1914535257821787      | 0.9473692520324072 | 0.036066892442986026 |
| <b>14.0</b> | 2.0   | 0.25     | 0.007099840930497378 | 0.01474572038219945     | 0.9485143736600773 | 0.014199681860994757 |

Figure 3: Effective dispersion relation derived from the Hamiltonian spectrum, exhibiting quadratic scaling consistent with Schrödinger-like dynamics.

## 5.6 Interpretation

These results indicate that wave-like dynamics naturally emerge from the graph structure.

In particular:

- the Laplacian governs spreading of the wavefunction,
- the degree-based potential introduces structural heterogeneity,
- the resulting dynamics exhibit features analogous to Schrödinger evolution.

Importantly, no continuum equation is imposed. The observed behaviour arises entirely from the discrete operator structure defined on the filament graph.

This provides evidence that quantum-like dynamics can arise as an effective regime of the underlying discrete system.

## 6 Emergent Dirac Structure

### 6.1 Dirac-like Operator on the Graph

Using the incidence matrix  $B$  introduced in Section 4, we define a Dirac-like operator:

$$D = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}. \quad (29)$$

This operator acts on the combined space of node and edge degrees of freedom.

### 6.2 Spectral Symmetry

The spectrum of  $D$  exhibits a characteristic symmetry:

$$E \leftrightarrow -E. \quad (30)$$

This symmetry arises directly from the block structure of the operator and implies that eigenvalues naturally organize into pairs  $(+E, -E)$ .

### 6.3 Zero Modes

We define zero modes as eigenvalues satisfying:

$$|E| < \varepsilon, \quad (31)$$

for a small threshold  $\varepsilon$ .

The number of zero modes reflects structural properties of the graph, including connectivity and the presence of cycles.

### 6.4 Spectral Pairing and Symmetry Error

To quantify the symmetry of the spectrum, we compare positive and negative eigenvalues:

$$\Delta = \frac{1}{N} \sum_k |E_k^+ - |E_k^-||. \quad (32)$$

Numerically, this symmetry error remains small across all tested configurations, confirming the robustness of the pairing structure.

## 6.5 Dispersion Relation

We analyse the low-energy part of the spectrum by defining:

$$k_{\text{eff}} = \sqrt{k}, \quad (33)$$

where  $k$  is the mode index.

Empirically, the eigenvalues follow a linear scaling:

$$E_k \sim v k_{\text{eff}}, \quad (34)$$

indicating a linear dispersion relation.

The coefficient  $v$  defines an emergent velocity scale within the model.

## 6.6 Localization Properties

We characterise eigenmodes using the effective support:

$$S_{\text{eff}} = \frac{1}{\sum_i |\psi_i|^4}. \quad (35)$$

Low-energy modes tend to exhibit extended support across the graph, while higher-energy modes may become more localized.

## 6.7 Interpretation

These results indicate that fermion-like spectral and dynamical features emerge naturally from the graph structure.

In particular:

- the symmetric spectrum reflects particle–antiparticle pairing,
- the linear dispersion is characteristic of relativistic dynamics,
- the emergent velocity defines a natural propagation scale.

Importantly, these properties arise without introducing spinors or continuum Dirac equations explicitly. They emerge from the combinatorial structure of the incidence matrix.

This suggests that fermion-like behaviour may be interpreted as an effective regime of the underlying discrete system.

# 7 Emergent Gauge Dynamics

## 7.1 Edge-Based Representation

Gauge-like dynamics naturally arise when considering fields defined on the edges of the filament graph.

Using the incidence matrix  $B$ , we define edge-based fields  $\psi \in \mathbb{R}^M$ , where  $M$  is the number of edges.

The incidence matrix provides discrete differential operators:

- $B^T$  acts as a discrete gradient,
- $B$  acts as a discrete divergence.

## 7.2 Edge Laplacian

The edge Laplacian is defined as:

$$L_e = B^T B. \quad (36)$$

This operator governs the dynamics of edge-based fields and encodes circulation and transverse structures.

## 7.3 Mode Decomposition

The eigenmodes of  $L_e$  can be classified into:

- zero modes corresponding to gradient fields (longitudinal),
- non-zero modes corresponding to transverse excitations.

The divergence of a mode  $\psi$  is given by:

$$\nabla \cdot \psi = B\psi. \quad (37)$$

For transverse modes, we expect:

$$\|B\psi\| \approx 0. \quad (38)$$

## 7.4 Dispersion Behaviour

The eigenvalues  $\lambda_k$  of  $L_e$  exhibit a scaling behaviour consistent with:

$$\lambda_k \sim k_{\text{eff}}^2, \quad (39)$$

where  $k_{\text{eff}} = \sqrt{k}$ .

Defining:

$$\omega_k = \sqrt{\lambda_k}, \quad (40)$$

we obtain:

$$\omega_k \sim k_{\text{eff}}, \quad (41)$$

indicating a linear dispersion relation.

## 7.5 Numerical Observations

Across all tested configurations:

- non-zero modes exhibit low divergence norms,
- transverse modes dominate the spectrum,
- dispersion remains stable across  $\beta$  and seeds.

## 7.6 Interpretation

These results indicate that gauge-like dynamics emerge naturally from the edge structure of the graph.

In particular:

- transverse propagation arises from divergence-free modes,
- linear dispersion is characteristic of wave propagation,
- the edge Laplacian plays a role analogous to the Maxwell operator.

Importantly, no continuum gauge field equations are imposed. The observed behaviour emerges entirely from the discrete graph structure.

This suggests that gauge-like dynamics may be interpreted as an effective regime of the underlying discrete system.

## 8 Emergent Geometry and Source Coupling

### 8.1 Collective Density Field

To probe large-scale organization of the filament graph, we construct a collective density field  $\rho(x)$  defined on the underlying lattice.

This field is obtained by aggregating shortest-path occupancies between selected high-degree nodes (hubs) of the graph. Each path contributes to the occupancy of the nodes it traverses.

The resulting discrete field is then smoothed and normalized:

$$\rho \leftarrow \frac{\mathcal{G}_\sigma * \rho}{\max(\rho)}, \quad (42)$$

where  $\mathcal{G}_\sigma$  denotes Gaussian smoothing.

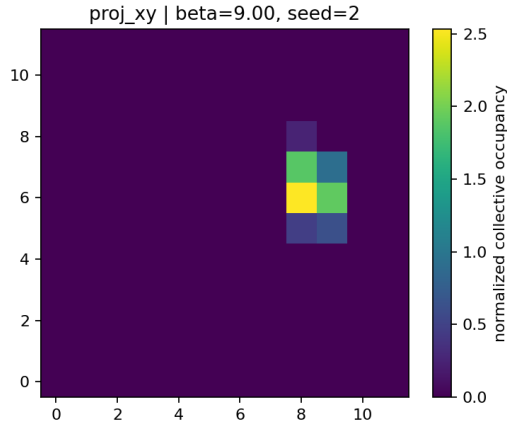


Figure 4: Projection of the normalized collective density field, showing localized structure emerging from path aggregation.

## 8.2 Differential Quantities

From the density field  $\rho$ , we define local quantities:

- gradient:

$$\nabla\rho, \tag{43}$$

- Laplacian:

$$\Delta\rho, \tag{44}$$

- Hessian tensor:

$$H_{ij} = \partial_i\partial_j\rho. \tag{45}$$

These quantities are computed using discrete finite-difference operators on the lattice.

## 8.3 Effective Metric

We define an effective metric tensor:

$$g_{ij} = \delta_{ij} + \alpha \partial_i\rho \partial_j\rho + \beta H_{ij}, \tag{46}$$

where  $\alpha$  and  $\beta$  are model parameters.

This construction captures:

- anisotropy induced by gradients,
- curvature-like effects encoded in second derivatives.

## 8.4 Effective Source Tensor

We define an effective source tensor:

$$T_{ij} = \partial_i\rho \partial_j\rho + \gamma \delta_{ij}|\nabla\rho|^2 + \eta \delta_{ij}\Delta\rho, \tag{47}$$

where  $\gamma$  and  $\eta$  control isotropic contributions.

## 8.5 Statistical Correlations

To assess the relationship between geometry and source, we compute correlations between:

- $\text{Tr}(T)$  and  $\text{Tr}(g)$ ,
- $\text{Tr}(T)$  and  $\det(g)$ ,
- anisotropy measures of  $T$  and  $g$ ,
- $\text{Tr}(T)$  and curvature proxy  $-\Delta\rho$ .

Across all tested values of  $\beta$  and independent seeds, we observe stable, non-trivial correlations.

## 8.6 Interpretation

These results indicate the emergence of a coupling between density-induced sources and geometric deformation.

In particular:

- the effective geometry responds to local density variations,
- the coupling is structured and robust across realizations,
- the behaviour is consistent with a geometry–source interaction.

Importantly, no continuum gravitational equations are imposed. The observed behaviour arises from the discrete organization of the system.

This suggests that geometry–source coupling may emerge as an effective regime of the underlying discrete dynamics.

## 8.7 Limitations

We emphasize that:

- the metric is effective and not derived from a variational principle,
- no direct equivalence with general relativity is claimed,
- physical units remain undefined in the present framework.

# 9 Minimal Dynamical Structure

## 9.1 Spectral Pair Dynamics

The Dirac-like operator introduced in Section 6 exhibits a symmetric spectrum, with eigenvalues naturally organized into pairs  $(+E, -E)$ .

To probe the intrinsic dynamics associated with this structure, we construct initial states from individual spectral pairs:

$$\psi_0^{(k)} = \frac{1}{\sqrt{2}} (\phi_{+k} + \phi_{-k}), \quad (48)$$

where  $\phi_{\pm k}$  are eigenvectors associated with eigenvalues  $\pm E_k$ .

The time evolution is given by:

$$\psi^{(k)}(t) = e^{-iDt} \psi_0^{(k)}. \quad (49)$$

## 9.2 Autocorrelation Function

We define the autocorrelation:

$$C_k(t) = \left| \langle \psi_0^{(k)}, \psi^{(k)}(t) \rangle \right|^2. \quad (50)$$

Numerically, this quantity exhibits highly regular oscillatory behaviour.

### 9.3 Observed Oscillatory Behaviour

Across all tested values of  $\beta$  and independent seeds:

- each spectral pair produces a dominant oscillation,
- the oscillation frequency matches  $2E_k$ ,
- deviations remain negligible.

This behaviour is consistent with a two-mode interference mechanism.

### 9.4 Interpretation

These observations indicate that the minimal dynamical structure associated with the Dirac-like operator is governed by pairwise oscillatory modes.

Importantly:

- the dynamics arise from individual spectral pairs,
- no additional intrinsic frequency scale is detected,
- the behaviour is fully determined by the operator spectrum.

This suggests that the fundamental temporal structure of the system, at this level of description, is controlled by simple two-mode interference.

### 9.5 Emergent Velocity Scale

The linear dispersion relation observed in Section 6 defines a natural velocity scale:

$$v = \frac{dE}{dk}. \quad (51)$$

This scale governs the propagation of modes in the system and provides a possible bridge toward physical calibration.

### 9.6 Position Within the Framework

This minimal dynamical structure represents the simplest non-trivial temporal behaviour supported by the model.

Higher-level structures observed in previous sections (wave propagation, gauge dynamics, geometry–source coupling) can be interpreted as collective or coarse-grained manifestations of these elementary spectral dynamics.

## 10 Unified Structural Formulation

### 10.1 Common Underlying Structure

The results presented in the previous sections indicate that multiple distinct dynamical regimes arise from a common underlying structure.

Starting from a scalar field defined on a discrete lattice, we extract filamentary structures that define a graph  $G = (V, E)$ . All subsequent operators and dynamics are constructed from this graph.

Importantly, no continuum equations are imposed at any stage. All observed behaviours emerge from the intrinsic properties of the discrete system.



## 10.2 Operator-Based Regimes

Different physical regimes correspond to different operators acting on the same underlying graph:

- Schrödinger-like dynamics arise from the normalized graph Laplacian and an effective potential,
- Dirac-like behaviour emerges from the incidence-based operator coupling nodes and edges,
- gauge-like dynamics are associated with the edge Laplacian acting on edge fields,
- geometry–source coupling arises from collective density fields defined on the lattice.

These regimes are not independent constructions, but rather different manifestations of the same discrete structure.

## 10.3 Hierarchy of Descriptions

The framework suggests a hierarchical organization:

- local variations of the scalar field define filamentary structures,
- these structures define a graph supporting discrete operators,
- operator spectra define elementary dynamical modes,
- collective behaviour of these modes gives rise to higher-level effective dynamics.

In this view, familiar physical equations correspond to effective descriptions of different levels of this hierarchy.

## 10.4 Minimal Dynamical Basis

At the most elementary level, the dynamics are governed by pairwise spectral interactions associated with the Dirac-like operator.

Higher-level phenomena observed in the framework can be interpreted as collective manifestations of these elementary dynamics.

This suggests that complex physical behaviour may arise from simple underlying spectral structures.

## 10.5 Minimal Generative Structure

The entire framework can be expressed as a sequence of transformations starting from a scalar field defined on a discrete lattice:

$$E : \Lambda \rightarrow \mathbb{R}. \quad (52)$$

From this field, all subsequent structures are derived deterministically through successive operations.

First, a critical field is constructed from local variations:

$$C(x) = \mathcal{S} \left( \max \left( 0, \frac{\|\nabla E(x)\| - \mu}{\sigma} \right)^2 \right), \quad (53)$$

where  $\mathcal{S}$  denotes a smoothing operator.

A thresholding operation defines a filament support:

$$\mathcal{F} = \{x \in \Lambda \mid C(x) \geq \theta\}. \quad (54)$$

From this support, a graph  $G = (V, E)$  is constructed using nearest-neighbour connectivity.

All dynamical regimes are then defined from operators acting on this graph:

$$H = \alpha L_{\text{norm}} + \beta V, \quad D = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad L_e = B^T B. \quad (55)$$

Finally, collective fields and effective geometric quantities are derived from graph-based observables:

$$\rho(x) = \mathcal{N} \left( \sum_{\gamma \in \Gamma} \chi_{\gamma}(x) \right). \quad (56)$$

This formulation highlights that all observed behaviours originate from a single scalar field through a sequence of structural transformations.

## 10.6 Collective Path Density Equation

A central component of the framework can be expressed through a compact relation that we refer to as the *Collective Path Density Equation*:

$$\rho(x) = \mathcal{N} \left( \sum_{\gamma \in \Gamma} \chi_{\gamma}(x) \right). \quad (57)$$

Here,  $\Gamma$  denotes a set of structurally relevant paths on the graph, and  $\chi_{\gamma}(x)$  is an indicator function describing the occupancy of node  $x$  by the path  $\gamma$ . The operator  $\mathcal{N}$  denotes normalization.

This equation expresses how collective structures emerge from the accumulation of path-based contributions on the graph. Regions where many paths intersect correspond to higher values of  $\rho(x)$ , giving rise to large-scale density patterns.

This compact formulation highlights the transition from discrete graph-based structures to continuous-like fields. It provides a bridge between the underlying combinatorial organization of the system and the emergent geometric and dynamical quantities discussed in previous sections.

## 10.7 Interpretation

The framework does not aim to replace established physical theories. Rather, it proposes an alternative perspective in which:

- different physical laws emerge from a common discrete substrate,
- continuum equations describe effective regimes of this substrate,
- the diversity of physical phenomena reflects different operator-based manifestations of the same underlying structure.

This interpretation is consistent with all observed numerical results and does not require introducing additional fundamental entities beyond the initial scalar field and its induced graph structure.

## 11 Code Availability and Reproducibility

### 11.1 Reference Implementation

A complete reference implementation of the framework is available online:

`https://github.com/NicolasGilbertAlbertRoux/unified-emergent-framework`

The repository provides all components required to reproduce the results presented in this work.

### 11.2 Code Structure

The implementation is organized into the following modules:

- `core/`: graph construction, operators, and utility functions,
- `physics/`: sector-specific analyses (Schrödinger, Dirac, Maxwell-like, Einstein-like),
- `preprocessing/`: generation of graph and density inputs,
- `configs/`: configuration files,
- `main_pipeline.py`: unified execution pipeline.

### 11.3 Input Data

The framework operates on scalar field configurations stored as NumPy arrays.

For reproducibility, the repository includes a lightweight dataset located in:

`data/raw_maps/`

This dataset is sufficient to reproduce all main results presented in the paper.

### 11.4 Execution Pipeline

The full analysis can be reproduced using the following steps:

1. Generate graph inputs:

```
python preprocessing/build_filament_graph_inputs.py
```

2. Generate density fields:

```
python preprocessing/build_collective_density_inputs.py
```

3. Run the unified pipeline:

```
python main_pipeline.py
```

All outputs are written to the `results/` directory.

## 11.5 Determinism

The pipeline is deterministic given fixed input data.

All stochasticity originates from the initial scalar field configurations. Once these inputs are fixed, all downstream computations are fully reproducible.

## 11.6 Scope of Reproducibility

The repository provides a computational reproduction of all numerical analyses presented in this work.

The framework is designed for transparency, verification, and extension by other researchers.

# 12 Discussion

## 12.1 Summary of Results

In this work, we have introduced a discrete computational framework in which multiple physical behaviours emerge from a common underlying structure.

Starting from scalar field configurations, we construct filamentary graphs that support several distinct dynamical regimes. These include:

- wave-like dynamics analogous to Schrödinger evolution,
- fermion-like spectral structures arising from a Dirac-type operator,
- gauge-like transverse modes derived from edge-based Laplacians,
- geometry–source coupling induced by collective density fields.

All these behaviours arise without imposing continuum equations, and are instead generated by the intrinsic properties of the discrete system.

## 12.2 Interpretation and Scope

The results suggest that different physical regimes may be interpreted as effective manifestations of a common underlying structure.

This work does not attempt to replace existing physical theories. Rather, it proposes a framework in which such theories may be viewed as emergent descriptions.

The approach is therefore complementary to established formulations, and aims to provide a unifying perspective rather than an alternative set of fundamental equations.

## 12.3 Limitations

Several limitations must be clearly acknowledged.

- The model is dimensionless and lacks direct physical calibration.
- No variational principle or action formulation is provided.
- The effective metric construction is heuristic and not derived from first principles.
- The correspondence with known physical theories remains qualitative.

In particular, no claim is made that the framework reproduces the full structure of quantum field theory or general relativity.

## 12.4 Relation to Existing Work

The idea that physical laws may emerge from discrete or combinatorial structures has been explored in various contexts, including lattice models, network-based approaches, and emergent gravity scenarios.

The present work contributes to this line of research by providing a unified computational framework in which several distinct physical regimes arise from the same underlying structure.

Rather than focusing on a single phenomenon, the framework simultaneously exhibits multiple behaviours typically associated with different domains of physics.

Several approaches have explored the emergence of physical laws from discrete or combinatorial structures, including lattice field theories, network-based models, and emergent gravity scenarios. The present work differs in that it provides a unified computational framework in which multiple physical regimes are observed simultaneously within a single structure.

## 12.5 Perspectives

Several directions for future work can be identified.

- introducing physical scales and calibration to connect the model with measurable quantities,
- exploring larger and more complex graph structures,
- investigating possible variational formulations,
- analysing stability and universality of the observed behaviours.

In addition, the framework may serve as a basis for further exploration of emergent descriptions of physical systems.

## 12.6 Final Remarks

The results presented here indicate that a single discrete structure can support a range of behaviours analogous to key domains of physics.

While the framework remains exploratory, it highlights the potential of emergent approaches, in which complex physical phenomena arise from simple underlying structures.

## 13 Conclusion

We have presented a discrete computational framework in which multiple physical behaviours emerge from a common underlying structure.

Starting from scalar field configurations defined on a lattice, filamentary structures are extracted and used to construct a graph. This graph supports a range of operators whose spectral properties give rise to distinct dynamical regimes.

These include wave-like propagation, fermion-like spectral pairing, gauge-like transverse modes, and geometry–source coupling. All these behaviours arise from the intrinsic properties of the discrete system, without imposing continuum equations.

The results suggest that diverse physical phenomena may be interpreted as effective manifestations of a shared underlying structure. In this perspective, well-established equations can be viewed as emergent descriptions of different regimes of a discrete substrate.

While the present framework remains exploratory and lacks direct physical calibration, it provides a unified computational setting in which multiple domains of physics can be investigated simultaneously.

This work highlights the potential of emergent approaches, where complex physical behaviour arises from simple underlying structures, and suggests several directions for further investigation.

## References

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## A Appendix A: Numerical Pipeline

The complete numerical workflow of the framework can be summarized as follows:

1. load a discrete scalar field  $E(x, y, z, t)$  from NumPy input arrays,
2. compute the gradient norm and identify high-variation regions,
3. extract filamentary supports via high-percentile thresholding,
4. construct a graph from nearest-neighbour filament connectivity,
5. build node- and edge-based operators,
6. evaluate the Schrödinger-like, Dirac-like, Maxwell-like, and geometry–source sectors,
7. compute derived observables and summary statistics.

## B Appendix B: Core Operators

The main operators used in the framework are:

### Normalized graph Hamiltonian

$$H = \alpha L_{\text{norm}} + \beta V$$

### Dirac-like operator

$$\mathcal{D} = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}$$

### Edge Laplacian

$$L_e = B^T B$$

### Effective metric

$$g_{ij} = \delta_{ij} + \alpha \partial_i \rho \partial_j \rho + \beta \partial_i \partial_j \rho$$

### Effective source tensor

$$T_{ij} = \partial_i \rho \partial_j \rho + \gamma \delta_{ij} |\nabla \rho|^2 + \eta \delta_{ij} \Delta \rho$$

## C Appendix C: Main Numerical Observables

The principal numerical observables used throughout the paper include:

- effective support

$$S_{\text{eff}} = \frac{1}{\sum_i |\psi_i|^4},$$

- Shannon entropy

$$H(\psi) = - \sum_i p_i \log p_i, \quad p_i = |\psi_i|^2,$$

- autocorrelation

$$C(t) = |\langle \psi_0, \psi(t) \rangle|^2,$$

- spectral pairing error in the Dirac-like sector,
- divergence norm in the Maxwell-like sector,
- correlation coefficients in the geometry–source sector.

## D Appendix D: Input and Output Data Structure

The framework relies on three main levels of data:

- raw input arrays in NumPy format,
- intermediate graph and density representations,
- sector-specific output summaries in CSV form.

Representative intermediate files include:

- filament node tables,
- filament edge tables,
- collective density arrays,
- spectral mode summaries.

## **E Appendix E: Code Availability**

The reference implementation associated with this work is publicly available at:

`https://github.com/NicolasGilbertAlbertRoux/unified-emergent-framework`

The repository contains the core graph utilities, physics modules, configuration files, preprocessing scripts, and the main execution pipeline.

## **F Appendix F: Interpretative Perspective**

The framework supports an interpretative perspective in which multiple effective physical regimes arise from a common discrete substrate.

Within this view, elementary oscillatory structures identified in the Dirac-like sector may serve as local organizational units. However, this interpretation remains heuristic at the present stage and is not required for the validity of the computational results reported in the main text.